Examiner only

3. In 1912, Alfred Wegener suggested that all of the continents on Earth were once joined together in one supercontinent called Pangaea. These continents have since drifted apart. Wegener suggested a hypothesis called ‘Continental Drift’. In the 1960’s scientists further developed Wegener’s hypothesis using a new theory called ‘Plate Tectonics’.

(a) (i) State **three** pieces of evidence that Wegener used to support his hypothesis of *‘Continental Drift’.* [3]

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(ii) Explain the theory of *‘Plate Tectonics’*. [3]

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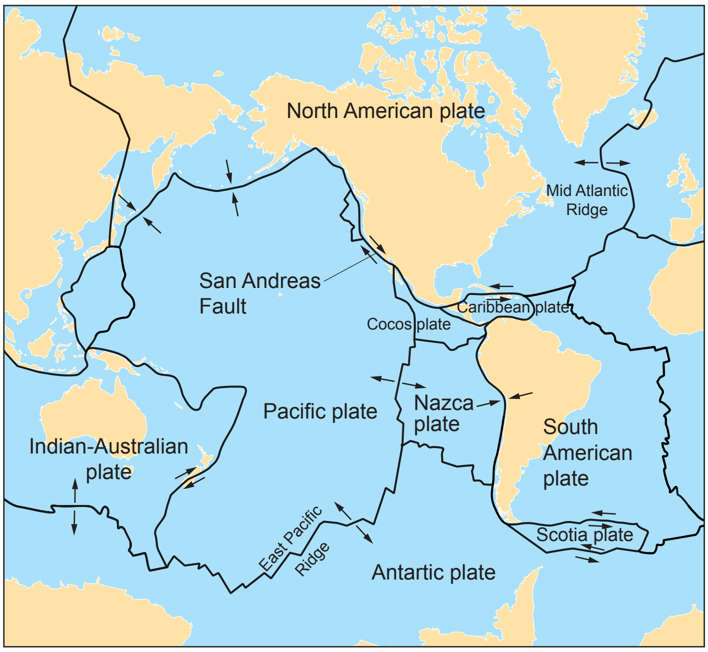
Examiner only

(b) The map below shows some of the tectonic plates on Earth and the arrows show the direction of their movement.



Tectonic plates move relative to each other and are monitored closely by geologists using a seismometer. The Pacific and North American plate are very closely monitored by the US government as large populations of people can be affected, especially in the state of California. The boundary between the Pacific and North American tectonic plates is described as a conservative boundary where the plates slide past each other causing powerful earthquakes. The San Andreas Fault lies on this boundary.

Tectonic plate boundaries can also be described as constructive and destructive.

(i) **Label** on the map **one** constructive (C) and **one** destructive (D) tectonic plate boundary. [1]

(ii) Describe these **two** types of boundary and the **consequences** of their movement.

Destructive .... [2]

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Constructive .... [2]

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